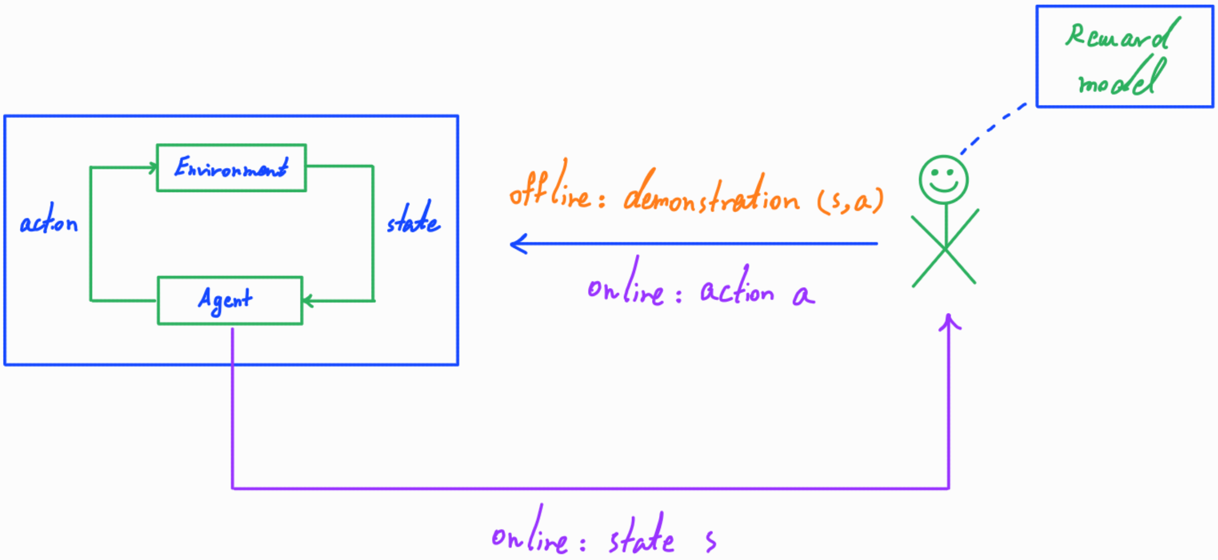
Imitation Learning

There are settings where

- a natural reward function does not exist,
- it is difficult to form a reward function, or
- it is difficult to learn from a reward function;

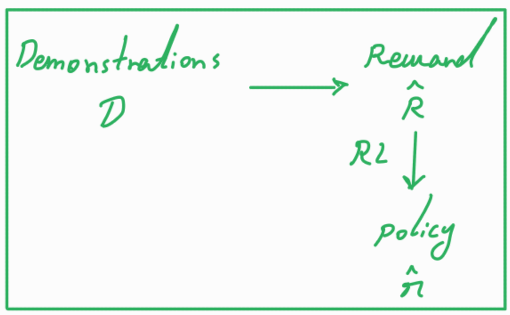
however, expert demonstrations (trajectories) are available.



In these settings, imitation learning provides a framework to learn a (near-) optimal policy from the demonstrations.

→ Behavior cloning (BC)

→ Inverse reinforcement learning (IRL)



Behavior cloning : BC aims to learn a policy directly from the demonstrations.

Consider a discounted, infinite-horizon MDP $\mathcal{M} = (\mathcal{S}, \mu_0, \mathcal{A}, P, R, \gamma)$ and an expert (near-optimal) policy π^*.

A data set of demonstrations from π^* over $\mathcal{M}$ is available

$$D = \{(s_i, a_i)\}_{i=1}^{N} \quad \text{where} \quad (s_i, a_i) \sim \bar{\nu}_{\mu_0}^{\pi^*}$$

The objective is to learn a policy from the data set D that is as good as the expert policy π^*.

This problem can be seen as supervised learning :

$$\hat{\pi} = \arg\min_{\pi \in \Pi} \mathbb{E}_{(s,a)\sim \bar{\nu}_{\mu_0}^{\pi^*}} \Big[\underbrace{L\big(\overbrace{\pi(\cdot|s)}^{\hat{a}}, a\big)}_{\text{Loss function}} \Big]$$

Candidate loss functions

- $\mathbb{1}\big[\pi(s_i) \neq \overbrace{a_i}^{\pi^*(s_i)}\big]$ discrete action
- $\big\| \pi(s_i) - \overbrace{a_i}^{\pi^*(s_i)} \big\|_2^2$ continuous action
- $-\log \pi(\underbrace{a_i}_{\pi^*(s_i)} | s_i)$ maximum likelihood estimation (MLE)

* If supervised learning works well, i.e., the empirical risk minimization (ERM) achieves near-optimal solution, then we have

$$\mathbb{E}_{(s,a)\sim \bar{\nu}_{\mu_0}^{\pi^*}} \left[L(\hat{\pi}(\cdot|s), a) \right] \leq \varepsilon \xrightarrow{} \text{e.g., } \frac{c}{\sqrt{N}}$$

Theorem: Assuming that a supervised learning algorithm yields $\hat{\pi}$ such that

$$\mathbb{E}_{(s,a)\sim \bar{\nu}_{\mu_0}^{\pi^*}} \left[\mathbb{1}\left[\hat{\pi}(s) \neq \pi^*(s) \right] \right] \leq \varepsilon,$$

we have the following sub-optimality bound:

$$V^{\pi^*}(s_0) - V^{\hat{\pi}}(s_0) \leq \frac{2R_{max}}{(1-\gamma)^2} \varepsilon \xrightarrow{} \text{e.g., } \frac{c}{\sqrt{N}}$$

Proof: From the performance difference lemma, it holds that

$$V^{\pi^*}(s_0) - V^{\hat{\pi}}(s_0) = \frac{1}{1-\gamma} \mathbb{E}_{s\sim \bar{\rho}_{\mu_0}^{\pi^*}} \left[\mathbb{E}_{a\sim \pi^*(\cdot|s)} \left[A^{\hat{\pi}}(s,a) \right] \right]$$

$A^{\pi}(s, \pi(s)) = 0$

$+$

$|R(s,a)| \leq R_{max} \Longrightarrow A^{\pi}(s,a) \leq \dfrac{2R_{max}}{1-\gamma}$

$$\leq \frac{1}{1-\gamma} \mathbb{E}_{s\sim \bar{\rho}_{\mu_0}^{\pi^*}} \left[\mathbb{E}_{a\sim \pi^*(\cdot|s)} \left[\frac{2R_{max}}{1-\gamma} \mathbb{1}\left[\hat{\pi}(s) \neq \pi^*(s) \right] \right] \right]$$

performance of supervised learning

$$\leq \frac{2R_{max}}{(1-\gamma)^2} \varepsilon.$$

Note: The performance of $\hat{\pi}$ will be potentially low in parts of the distribution not covered in the data set.

Data set Aggregation (DAgger): DAgger interacts with the environment and the expert to query new data to be added to the existing data set.

DAgger Algorithm

- Initialize policy π_0 and data set $D = \emptyset$
- For $t = 0, 1, 2, \dots, T-1$
  - Generate a data set with π_t and query from π^*

    $$D_t = \{s_i, a_i\}_{i=1}^{N} \quad \text{where} \quad s_i \sim \bar{\rho}_{\mu_0}^{\pi_t} \quad \text{and} \quad a_i \sim \pi^*(s_i)$$

  - Aggregate data set

    $$D \leftarrow D \cup D_t$$

  - Update policy by supervised learning

    $$\pi_{t+1} = \arg\min_{\pi \in \Pi} \sum_{(s,a) \in D} L(\pi(\cdot|s), a)$$

entropy of p_θ

regularizer, e.g., $-\mathcal{H}(\theta)$

$$\underbrace{\theta_{t+1} = \arg\min_{\theta \in \Theta} \left[\sum_{i=1}^{t} L_i + \lambda F(\theta) \right]}$$

Note: DAgger can be viewed as the follow-the-regularized-leader (FTRL) (FTRL) algorithm that under certain conditions achieves

$$\exists t \text{ such that } V^{\pi^*}(s_0) - V^{\pi_t}(s_0) \leq \frac{\max_{s,a} |A^{\pi^*}(s,a)|}{1-\gamma} \varepsilon .$$

$\leq \frac{2 R_{max}}{1-\gamma}$

e.g., $\frac{c}{\sqrt{N}}$